

ISEN/DAEN 427 Lecture 2

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Bayesian Inference and Estimation

BM Ch. 3, 4



Comparison between generative model and inference

	Generative model	Inference (decision model)
Point of view of	the world	the observer (decision-maker)
Is realized	across trials	on any single trial
Prior, likelihood, and posterior	play no role	are central
Nature of distributions	objective (probability of occurrence)	subjective (degrees of belief)
Outcome	expressions for the distributions of all variables, including the observations	a rule to map observations to a decision (estimate of world state)

Inference with Discrete States: Example

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- The distribution of occupations x is:

x	$s = 1$	$s = 2$	$s = 3$
Total Population	19771(79.0%)	3446(13.8%)	1805(7.2%)
teacher	297(1.5%)	28(0.8%)	20(1.1%)
farmer	79(0.4%)	90(2.6%)	65(3.6%)
retail	1661(8.4%)	320(9.3%)	150(8.3%)
other	17735(89.7%)	3008(87.3%)	1570(87.0%)

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- The likelihood of occupations $\mathbb{P}(x|s)$ is:

$\mathbb{P}(x s)$	$s = 1$	$s = 2$	$s = 3$
$x = \text{teacher}$	$\frac{297}{19771}$	$\frac{28}{3446}$	$\frac{20}{1805}$
$x = \text{farmer}$	$\frac{79}{19771}$	$\frac{90}{3446}$	$\frac{65}{1805}$
$x = \text{retail}$	$\frac{1661}{19771}$	$\frac{320}{3446}$	$\frac{150}{1805}$
$x = \text{other}$	$\frac{17735}{19771}$	$\frac{3008}{3446}$	$\frac{1570}{1805}$

Inference with Discrete States: Example

- The prior distribution is

	$s = 1$	$s = 2$	$s = 3$
$\mathbb{P}(s)$	$\frac{19771}{19771+3446+1805}$	$\frac{3446}{19771+3446+1805}$	$\frac{1805}{19771+3446+1805}$

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$\mathbb{P}(s)$	$\frac{19771}{19771+3446+1805}$	$\frac{3446}{19771+3446+1805}$	$\frac{1805}{19771+3446+1805}$

- The posterior distribution is:

$\mathbb{P}(s x)$	$s = 1$	$s = 2$	$s = 3$
$x = \text{teacher}$	0.86	0.07	0.05
$x = \text{farmer}$	0.33	0.38	0.27
$x = \text{retail}$	0.78	0.15	0.07

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- The prior distribution is Normal with mean μ_s and variance σ_s^2 :

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- The likelihood function is:

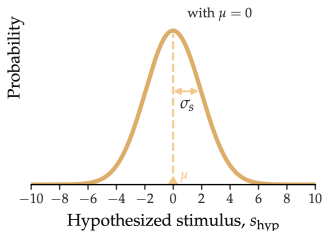
$$\mathbb{P}(x|s) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - s)^2}{2\sigma^2}\right)$$

Here the difference $x - s$ captures how far the observed measurement is from the hypothesized true state. If x is close to s , the exponent is near zero $\rightarrow$ high likelihood. If x is far from s , the exponent is very negative $\rightarrow$ low likelihood.

Inference

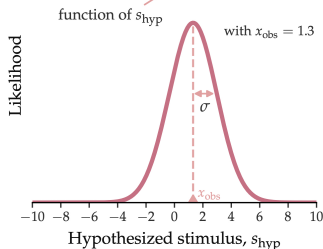
(A) Prior distribution

$$p_s(s_{\text{hyp}}) = \frac{1}{\sqrt{2\pi\sigma_s^2}} e^{-\frac{(s_{\text{hyp}} - \mu)^2}{2\sigma_s^2}}$$



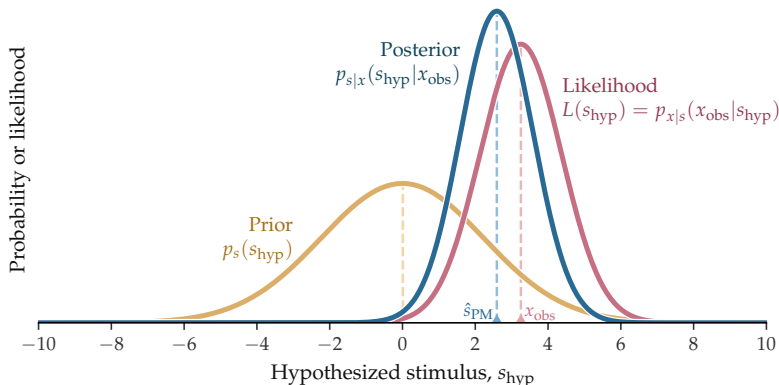
(B) Likelihood function

$$\mathcal{L}(s_{\text{hyp}}; x_{\text{obs}}) = p_{x|s}(x_{\text{obs}}|s_{\text{hyp}}) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_{\text{obs}} - s_{\text{hyp}})^2}{2\sigma^2}}$$



Consider a single trial on which the measurement is x_{obs} . The observer is trying to infer which value of the stimulus s produced this measurement. The two functions that play a role in the observer's inference process (on a single trial) are the prior and the likelihood. (A) Prior distribution reflects the observer's beliefs about different possible values the stimulus can take. (B) The likelihood function over the stimulus based on the measurement x_{obs} . The likelihood function is centered at x_{obs} .

Inference



The posterior distribution is obtained by multiplying the prior distribution with the likelihood function, and normalizing the resulting function.

- The *a posteriori* distribution is proportional to the product of the *likelihood* and *a priori* distributions:

$$\mathbb{P}(s|x) = \kappa \times \mathbb{P}(x|s)\mathbb{P}(s)$$

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- As before $\int \mathbb{P}(s|x)ds = 1$ implies

$$\kappa = \frac{1}{\int \mathbb{P}(x|s)\mathbb{P}(s)ds} = \frac{1}{\mathbb{P}(x)}$$

- Hence

$$\mathbb{P}(s|x) = \frac{\kappa}{\sqrt{2\pi\sigma}\sqrt{2\pi\sigma_s}} \exp\left(-\frac{(x-s)^2}{2\sigma^2} - \frac{(s-\mu_s)^2}{2\sigma_s^2}\right)$$

Posterior Distribution

- Hence

$$\mathbb{P}(s|x) = \frac{\kappa}{\sqrt{2\pi\sigma}\sqrt{2\pi\sigma_s}} \exp\left(-\frac{(x-s)^2}{2\sigma^2} - \frac{(s-\mu_s)^2}{2\sigma_s^2}\right)$$

- After a bit of algebra, it can be shown that the *a posteriori* distribution is also Gaussian with mean μ_{post} and variance σ_{post}^2 :

$$\mu_{post} = \frac{\frac{1}{\sigma^2}}{\frac{1}{\sigma^2} + \frac{1}{\sigma_s^2}} x + \frac{\frac{1}{\sigma_s^2}}{\frac{1}{\sigma^2} + \frac{1}{\sigma_s^2}} \mu_s \quad \sigma_{post}^2 = \frac{1}{\frac{1}{\sigma^2} + \frac{1}{\sigma_s^2}}$$

- Another useful way to describe the posterior distribution is in terms of *precision*:

$$J = \frac{1}{\sigma^2} \qquad J_s = \frac{1}{\sigma_s^2}$$

Posterior Distribution

- Another useful way to describe the posterior distribution is in terms of *precision*:

$$J = \frac{1}{\sigma^2} \quad J_s = \frac{1}{\sigma_s^2}$$

- The posterior distribution is Gaussian with mean and variance:

$$\mu_{post} = \frac{J}{J + J_s} \mathbf{x} + \frac{J_s}{J + J_s} \mu_s \quad \sigma_{post}^2 = \frac{1}{J + J_s}$$

- *Prior* uncertainty reflects the quality of the observer's knowledge about the state of the world before making any observations,

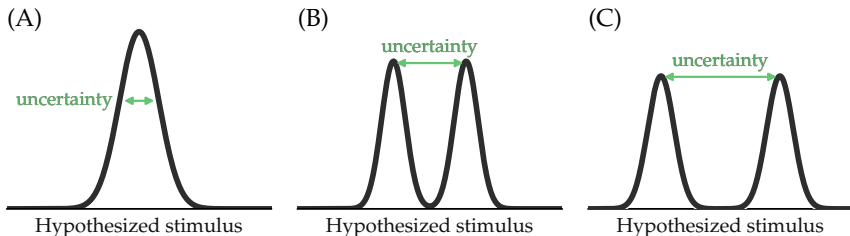
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- *Posterior* uncertainty reflects the quality of the observer's knowledge about the state of the world after making the observations.

Uncertainty

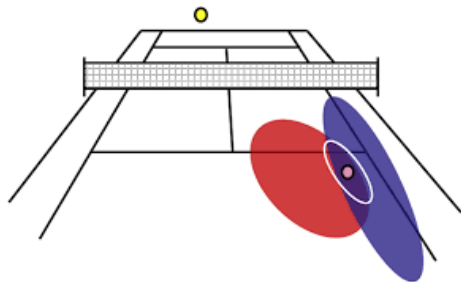


Increasing uncertainty, defined as the standard deviation of a random variable. Each of the distributions shown could be the prior, the (normalized) likelihood, or the posterior. The double arrow is two standard deviations long. (B) and (C) show bimodal (two-peaked) distributions. Uncertainty in (C) is higher solely due to the greater separation between the peaks.

	Mean	Variance
Step 1: Generative model	Stimulus distribution, $p(s)$ μ Measurement distribution, $p(x s)$ s	σ_s^2 σ^2
Step 2: Inference	Prior distribution, $p(s)$ μ Likelihood function $\mathcal{L}(s_{\text{hyp}}; x_{\text{obs}}) = p(x_{\text{obs}} s_{\text{hyp}})$ x_{obs} Posterior distribution $p(s_{\text{hyp}} x_{\text{obs}})$ $\hat{s}_{\text{PM}} = \frac{Jx_{\text{obs}} + J_s\mu}{J + J_s}$	σ_s^2 σ^2 $\frac{1}{J + J_s}$
Step 3: Response distribution	Response distribution, $p(\hat{s} s)$ $\mathbb{E}[\hat{s}_{\text{PM}} s] = \frac{J_s + J_s\mu}{J + J_s}$	$\frac{J}{(J + J_s)^2}$

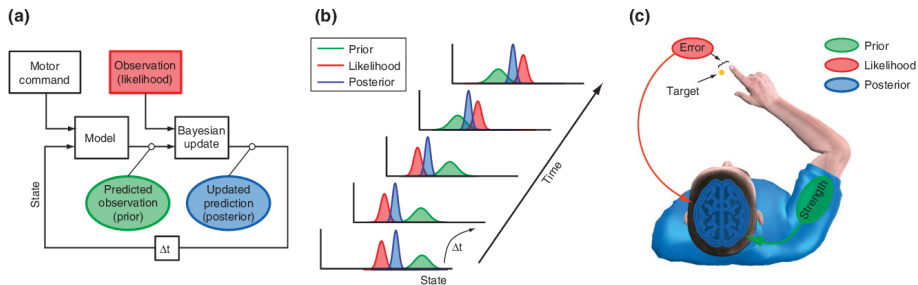
Applications: Visual Perception

- The prior is shown in **blue**, the likelihood distribution in **red**, and the posterior distribution with the *white* ellipse.
- The maximum posterior estimate is shown by the **magenta** ball.
- This estimate can be updated in light of new information about the ball's trajectory (yellow).



Bayesian Inference

Applications: Human Sensorimotor Integration



Motor adaptation: prior belief in muscle properties (e.g., muscle strength) in green is integrated with our observed motor errors in red to update estimates in our muscle properties blue.